1. ABSTRACT

The **Robotics Control System** is implemented using a **Graph data structure (Adjacency List)** to manage communication between sensors efficiently.
It supports operations such as:

- Adding connections (edges)
- Deleting nodes
- Searching nodes
- Updating node values
- Displaying the network

The graph structure enables efficient representation of sensor communication networks and demonstrates real-world applications of data structures in robotics.

---

2. INTRODUCTION

Modern robotic systems rely on multiple sensors that communicate with each other.
Efficient data handling is crucial for system performance.

Traditional storage methods are inefficient for representing connections.
A **Graph data structure** provides:

- Flexible relationships between nodes
- Efficient traversal and updates
- Clear representation of communication networks

Thus, graphs are highly suitable for robotics control systems.

---

3. OBJECTIVES

- To design a system for managing sensor communication
- To implement graph operations like add, delete, search, and update
- To represent real-time connections between sensors
- To improve efficiency over traditional data structures
- To understand real-world applications of graphs

---

4. SCOPE OF THE PROJECT

- Can be used in robotics sensor networks
- Suitable for communication mapping systems
- Helps in monitoring and updating connections quickly
- Can be extended to AI-based robotic systems
- Useful in IOT and automation networks

---

5. LITERATURE REVIEW

- Graphs are widely used for network representation
- Adjacency list provides space-efficient storage
- Time complexity:
  - Insertion: O(1)
  - Search: O(V + E)
- Used in:
  - Computer networks
  - Robotics systems
  - Navigation systems

Limitations:

- Traversal can be complex for large graphs

Advanced Concepts:

- Directed and Undirected Graphs
- Weighted Graphs
- Graph Traversal Algorithms (DFS, BFS)

---

6. METHODOLOGY

The system is implemented using a **menu-driven approach**:

Steps:

1. Create graph using dynamic memory
2. Add connection between sensors
3. Search for a node in the network
4. Update node values

5. Delete nodes from the graph
6. Display graph using adjacency list

Each operation is selected by the user and executed accordingly.

---

7. TECHNOLOGY USED

- **Programming Language:** C
- **Concepts Used:**
  - Graph (Adjacency List)
  - Pointers
  - Dynamic Memory Allocation
  - Structures
  - Functions

---

8. ARCHITECTURE / SYSTEM DESIGN

Structure:

- Graph contains:
  - Number of vertices (sensors)
  - Array of linked lists
- Each node contains:
  - Sensor ID (data)
  - Pointer to next node

Flow:

User Input → Menu → Function Call → Graph Operation → Output Display

---

9. IMPLEMENTATION DETAILS

The system uses the following functions:

- `CreateNode()` → Creates a new sensor node
- `createGraph()` → Initializes graph
- `addNode()` → Adds communication link
- `deleteNode()` → Removes node from network

- `updateNode()` → Updates node value
- `searchNode()` → Searches node in graph
- `displayGraph()` → Displays network

The program uses:

- Loops for traversal
- Linked lists for adjacency representation
- Switch-case for menu handling

---

10. RESULTS / OUTPUT

The system successfully performs:

- Adding sensor connections
- Searching nodes
- Updating node values
- Deleting nodes
- Displaying communication network

Sample Output:

```
===== Robotics Control Menu =====
 1. Add Node
 2. Delete Node
 3. Update Node
 4. Search
 5. Display
 6. Exit
```

---

11. CONCLUSION

The **Graph-based Robotics Control System**:

- Efficiently manages sensor communication
- Provides flexible and dynamic connections
- Supports all key operations effectively

Advantages:

- Easy to implement
- Scalable for network-based systems
- Efficient for representing relationships

12. FUTURE WORK

- Implement **weighted graphs** for distance-based communication
- Add **DFS/BFS traversal algorithms**
- Integrate with real-time robotic hardware
- Add **GUI for visualization**
- Use databases for persistent storage
- Extend to IOT-based smart systems
- Develop web/mobile-based monitoring system

13. REFERENCES

- Data Structures using C textbooks
- Online tutorials on Graphs
- Geeks for Geeks (Graph concepts)
- Classroom notes and lecture materials